

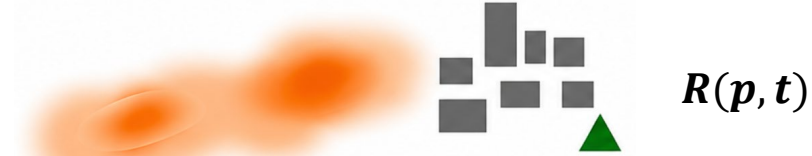
1. Mobility Cost from Density



high density $\rightarrow$ low speed $\rightarrow$ high cost

$\times \alpha$

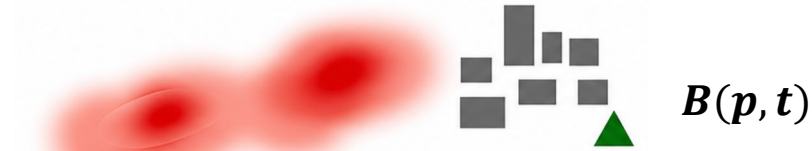
2. Continuous Hazard Cost (R)



hazard proximity / intensity $\rightarrow$ risk penalty

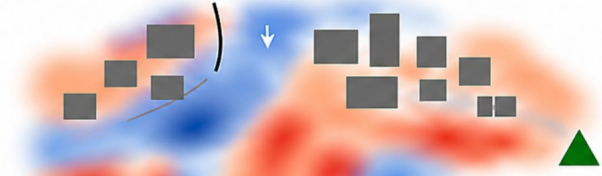
$\times \beta$

3. Hard Blocking Region (B)



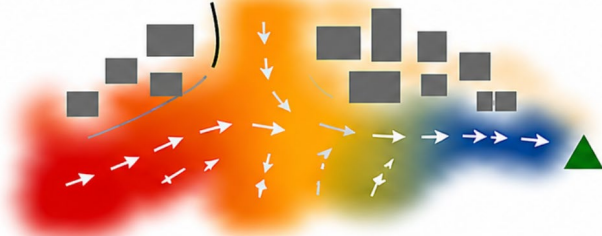
blocked cells $\rightarrow$ impassable

$\times \gamma$



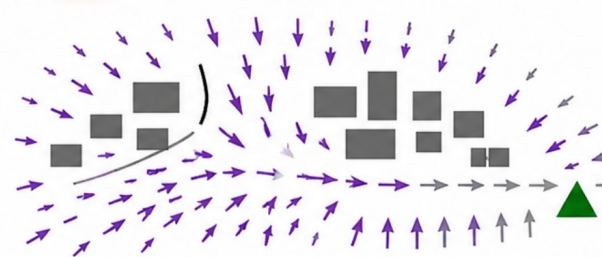
Weighted Composite Cost Field

$$C(p, t) = \alpha C_p(p, t) + \beta R(p, t) + \gamma B(p, t)$$



Cumulative Navigation Potential

$$\Phi(p, t) = \min_{\pi: p \rightarrow \Omega_{exit}} \int_{\pi} C(s, t) ds$$



Global Navigation Direction

$$g(p, t) = -\frac{\nabla \Phi(p, t)}{\|\nabla \Phi(p, t)\| + \varepsilon}$$